

Laplace Transform

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

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MATHEMATICS FOR INFORMATION TECHNOLOGY

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Outline



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Matrices
 - The Inverse of a Matrix
- Determinants
 - The Determinant of a Matrix
 - Applications of Determinants
- Applications of Matrix Operations
 - Cryptography

The Inverse of a Matrix

A decorative image in the top right corner showing a portion of a document with the Laplace Transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

An $n \times n$ matrix A is **invertible** (or **nonsingular**) if there exists an $n \times n$ matrix B such that

$$AB = BA = I_n$$

where I_n is the identity matrix of order n . The matrix B is called the (multiplicative) **inverse** of A . A matrix that does not have an inverse is called **noninvertible** (or **singular**).

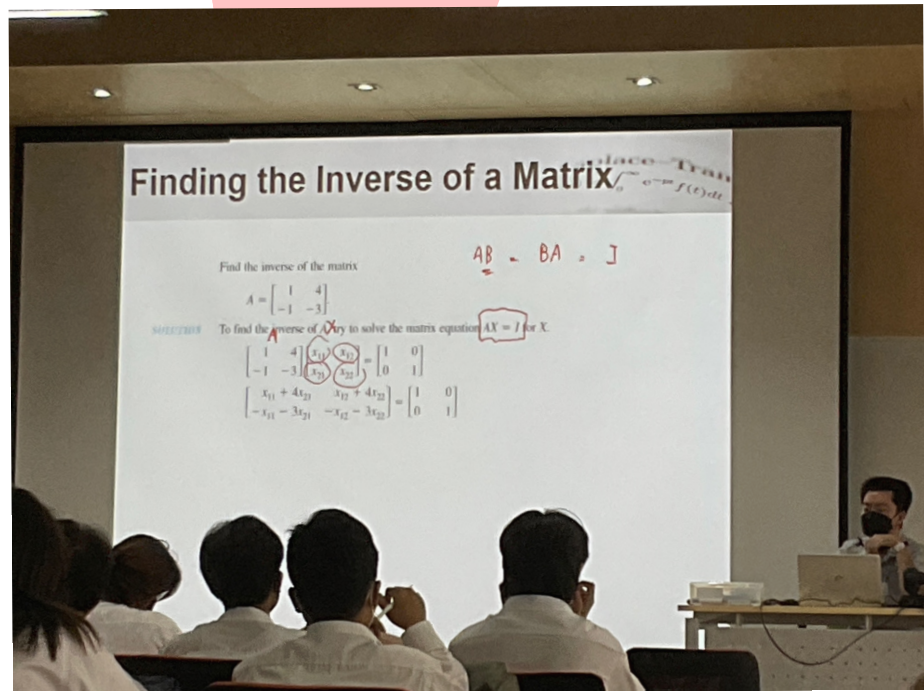
Finding the Inverse of a Matrix

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 4 \\ -1 & -3 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\frac{1}{-3+4} \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} -3 & -4 \\ 1 & 1 \end{bmatrix}$$



Properties of Inverse Matrices

A decorative image in the top right corner showing a blurred snippet of a document with the Laplace transform formula: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.

If A is an invertible matrix, k is a positive integer, and c is a scalar not equal to zero, then A^{-1} , A^k , cA , and A^T are invertible and the following are true.

1. $(A^{-1})^{-1} = A$
2. $(A^k)^{-1} = A^{-1}A^{-1} \cdot \cdot \cdot A^{-1} = (A^{-1})^k$
3. $(cA)^{-1} = \frac{1}{c}A^{-1}, c \neq 0$
4. $(A^T)^{-1} = (A^{-1})^T$

If A and B are invertible matrices of size n , then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}.$$

The Determinant of a Matrix

- Every *square* matrix can be associated with a real number called its **determinant**.
- Historically, the use of determinants arose from the recognition of special patterns that occur in the solutions of systems of linear equations.

Definition of the Determinant of a 2×2 Matrix

The **determinant** of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

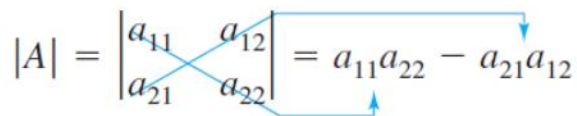
is given by

$$\det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}.$$

REMARK: In this text, $\det(A)$ and $|A|$ are used interchangeably to represent the determinant of a matrix. Vertical bars are also used to denote the absolute value of a real number; the context will show which use is intended. Furthermore, it is common practice to delete the matrix brackets and write

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \quad \text{instead of} \quad \left| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right|.$$

A convenient method for remembering the formula for the determinant of a 2×2 matrix is shown in the diagram below.

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$


The determinant is the difference of the products of the two diagonals of the matrix. Note that the order is important, as demonstrated above.

The Determinant of a Matrix of Order 2

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$
with surface

Find the determinant of each matrix.

(a) $A = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$

$$4 + 3 = 7$$

(b) $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$

$$4 - 4 = 0$$

(c) $C = \begin{bmatrix} 0 & 3 \\ 2 & 4 \end{bmatrix}$

$$0 - 6 = -6$$

Definitions of Minors and Cofactors of a Matrix

Laplace-Trans

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix, then the **minor** M_{ij} of the element a_{ij} is the determinant of the matrix obtained by deleting the i th row and j th column of A . The **cofactor** C_{ij} is given by

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

For example, if A is a 3×3 matrix, then the minors and cofactors of a_{21} and a_{22} are as shown in the diagram below.

Minor of a_{21}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{21} = \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}$$

Delete row 2 and column 1.

Cofactor of a_{21}

$$\begin{aligned} C_{21} &= (-1)^{2+1} M_{21} \\ &= -M_{21} \end{aligned}$$

Minor of a_{22}

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad M_{22} = \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

Delete row 2 and column 2.

Cofactor of a_{22}

$$\begin{aligned} C_{22} &= (-1)^{2+2} M_{22} \\ &= M_{22} \end{aligned}$$

Definitions of Minors and Cofactors of a Matrix (cont.)

As you can see, the minors and cofactors of a matrix can differ only in sign. To obtain the cofactors of a matrix, first find the minors and then apply the checkerboard pattern of +’s and –’s shown below.

Sign Pattern for Cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

3 × 3 matrix

$$\begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix}$$

4 × 4 matrix

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

n × n matrix

Note that *odd* positions (where $i + j$ is odd) have negative signs, and even positions (where $i + j$ is even) have positive signs.

Find the Minors and Cofactors of a Matrix

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Find all the minors and cofactors of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} = -1$$

$$M_{12} = \begin{bmatrix} 3 & 2 \\ 4 & 1 \end{bmatrix} = -5$$

$$M_{13} = \begin{bmatrix} 3 & -1 \\ 4 & 0 \end{bmatrix} = 4$$

$$M_{21} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = 2$$

$$M_{22} = \begin{bmatrix} 0 & 1 \\ 4 & 1 \end{bmatrix} = -4$$

$$M_{23} = \begin{bmatrix} 0 & 2 \\ 4 & 0 \end{bmatrix} = -8$$

$$M_{31} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} = 5$$

$$M_{32} = \begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} = -3$$

$$M_{33} = \begin{bmatrix} 0 & 2 \\ 3 & -1 \end{bmatrix} = -6$$

$$C_{11} = -1 - 0 = -1$$

$$C_{12} = -(3 - 8) = 5$$

$$C_{13} = 0 + 4 = 4$$

$$C_{21} = -(2 - 0) = -2$$

$$C_{22} = 0 - 4 = -4$$

$$C_{23} = -(0 - 8) = 8$$

$$C_{31} = 4 + 1 = 5$$

$$C_{32} = -(0 - 3) = 3$$

$$C_{33} = 0 - 6 = -6$$

Definition of the Determinant of a Matrix

A decorative image in the top right corner showing a blurred mathematical formula for the Laplace transform: $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$.
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt$$

If A is a square matrix (of order 2 or greater), then the determinant of A is the sum of the entries in the first row of A multiplied by their cofactors. That is,

$$\det(A) = |A| = \sum_{j=1}^n a_{1j}C_{1j} = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}.$$

The Determinant of a Matrix of Order 3

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

6/8/2020
Handwritten notes and arrows indicating the expansion process.

$$|A| = 0(-1) + 2(5) + 1(4)$$

$$= 14$$

$$|A| = 3(-2) + (-1)(-4) + 2(8)$$

$$= 14$$

$$|A| = 4(5) + 0(3) + 1(-6)$$

$$= 14$$

Expansion by Cofactors



Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Let A be a square matrix of order n . Then the determinant of A is given by

$$\det(A) = |A| = \sum_{j=1}^n a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

or

$$\det(A) = |A| = \sum_{i=1}^n a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}.$$

The Determinant of a Matrix of Order 4

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Find the determinant of

$$A = \begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix}.$$

SOLUTION

By inspecting this matrix, you can see that three of the entries in the third column are zeros. You can eliminate some of the work in the expansion by using the third column.

$$|A| = 3(C_{13}) + 0(C_{23}) + 0(C_{33}) + 0(C_{43})$$

Because C_{23} , C_{33} , and C_{43} have zero coefficients, you need only find the cofactor C_{13} . To do this, delete the first row and third column of A and evaluate the determinant of the resulting matrix.

$$C_{13} = (-1)^{1+3} \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix}$$

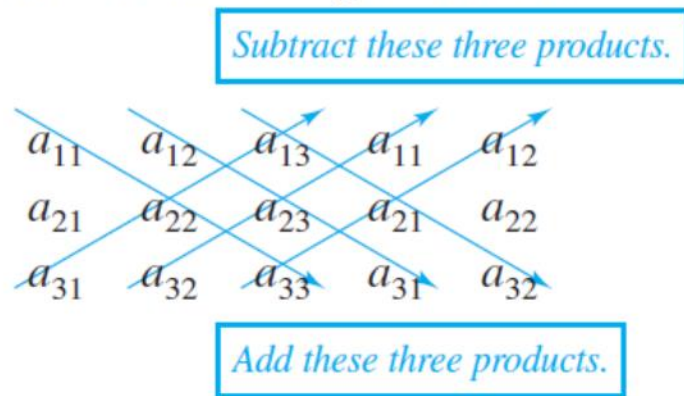
Expanding by cofactors in the second row yields

$$\begin{aligned} C_{13} &= (0)(-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix} + (2)(-1)^{2+2} \begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix} + (3)(-1)^{2+3} \begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix} \\ &= 0 + 2(1)(-4) + 3(-1)(-7) = 13. \end{aligned}$$

You obtain $|A| = 3(13) = 39$.

Definition of the Determinant of a Matrix (cont.)

There is an alternative method commonly used for evaluating the determinant of a 3×3 matrix A . To apply this method, copy the first and second columns of A to form fourth and fifth columns. The determinant of A is then obtained by adding (or subtracting) the products of the six diagonals, as shown in the following diagram.



Try confirming that the determinant of A is

$$|A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

The Determinant of a Matrix of Order 3

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Find the determinant of

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix} \begin{matrix} 0 & 2 \\ 3 & -1 \\ 4 & -4 \end{matrix}$$

$$= 0 + 16 - 12 + 4 - 0 - 6 = 2$$

Conditions That Yield a Zero Determinant

Laplace-Trans
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

If A is a square matrix and any one of the following conditions is true, then $\det(A) = 0$.

1. An entire row (or an entire column) consists of zeros.
2. Two rows (or columns) are equal.
3. One row (or column) is a multiple of another row (or column).

$$\begin{vmatrix} 0 & 0 & 0 \\ 2 & 4 & -5 \\ 3 & -5 & 2 \end{vmatrix} = 0,$$

The first row
has all zeros.

$$\begin{vmatrix} 1 & -2 & 4 \\ 0 & 1 & 2 \\ 1 & -2 & 4 \end{vmatrix} = 0,$$

The first and third
rows are the same.

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & -1 & -6 \\ -2 & 0 & 6 \end{vmatrix} = 0.$$

The third column is a
multiple of the first column.

Properties of Determinants

If A and B are square matrices of order n , then

$$\det(AB) = \det(A) \det(B).$$

If A is an $n \times n$ matrix and c is a scalar, then the determinant of cA is given by

$$\det(cA) = c^n \det(A).$$

A square matrix A is invertible (nonsingular) if and only if

$$\det(A) \neq 0.$$

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

If A is a square matrix, then

$$\det(A) = \det(A^T).$$

The Adjoint of a Matrix

Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- If A is a square matrix, then the **matrix of cofactors** of A has the form

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}.$$

- The transpose of this matrix is called the **adjoint** of A and is denoted by $\text{adj}(A)$. That is,

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}.$$

Finding the Adjoint of a Square Matrix

Laplace-Transform
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt$

Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 5 & 12 \\ 6 & 3 \\ 7 & 2 \end{bmatrix}$$

The Inverse of a Matrix Given by Its Adjoint

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

Using the Adjoint of a Matrix to Find Its Inverse

Use the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

to find A^{-1} .

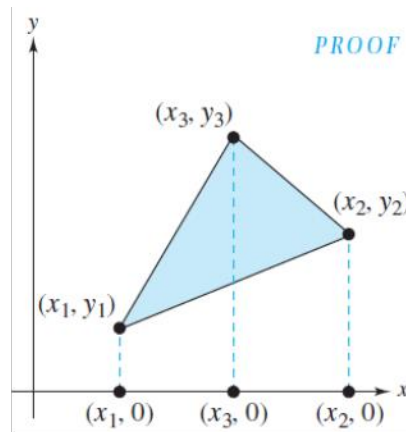
$$\frac{1}{3} \cdot \begin{bmatrix} 6 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}$$

Applications of Determinants

- Area, Volume, and Equations of Lines and Planes
- Two-Point Form of the Equation of a Line
- Volume of a Tetrahedron

Area, Volume, and Equations of Lines and Planes

$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



The area of the triangle whose vertices are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) is given by

$$\text{Area} = \pm \frac{1}{2} \det \begin{bmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive area.

Finding the Area of a Triangle

Find the area of the triangle whose vertices are $(1, 0)$, $(2, 2)$, and $(4, 3)$.

$$\frac{1}{2} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 1 \\ 4 & 3 & 1 \end{bmatrix} = -\frac{3}{2}$$

↓
 $\det = -3$

Two-Point Form of the Equation of a Line



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

An equation of the line passing through the distinct points (x_1, y_1) and (x_2, y_2) is given by

$$\det \begin{bmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{bmatrix} = 0.$$

Finding an Equation of the Line Passing Through Two Points

Find an equation of the line passing through the points $(2, 4)$ and $(-1, 3)$.

$$\begin{bmatrix} x & y & 1 \\ 2 & 4 & 1 \\ -1 & 3 & 1 \end{bmatrix} = 0$$

$$x \begin{bmatrix} 4 & 1 \\ 3 & 1 \end{bmatrix} - y \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + 1 \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix} = 0$$

$$x - 3y + 10 = 0$$

$$\text{Simplify} \Rightarrow x - 3y = -10$$

Volume of a Tetrahedron

$$\text{Laplace-Trans} \\ F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- เตตระฮีดรอน ตรงกับภาษาอังกฤษว่า Tetrahedron (อ่านว่า tet-tra-he-dron) หมายถึงโครงสร้างปิระมิดฐานสามเหลี่ยม ซึ่งประกอบด้วย ด้าน (face) สามเหลี่ยม 4 ด้าน มุม (vertex) 4 มุม ขอบ (edge) 6 ขอบ เตตระฮีดรอนเป็นรูปทรงที่จัดอยู่ในประเภท พลาโตนิก (platonic)

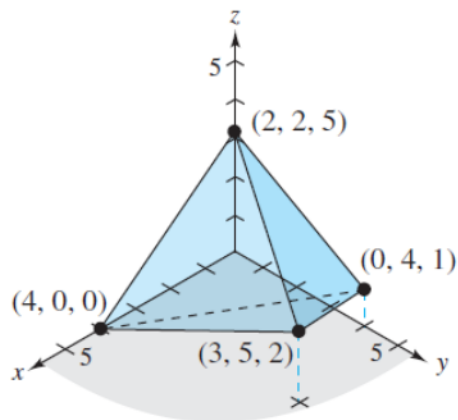
The volume of the tetrahedron whose vertices are (x_1, y_1, z_1) , (x_2, y_2, z_2) , (x_3, y_3, z_3) , and (x_4, y_4, z_4) is given by

$$\text{Volume} = \pm \frac{1}{6} \det \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{bmatrix},$$

where the sign $(\pm)$ is chosen to yield a positive volume.

Finding the Volume of a Tetrahedron

Find the volume of the tetrahedron whose vertices are $(0, 4, 1)$, $(4, 0, 0)$, $(3, 5, 2)$, and $(2, 2, 5)$



$$\frac{1}{6} \begin{bmatrix} 0 & 4 & 1 & 1 \\ 4 & 0 & 0 & 1 \\ 3 & 5 & 2 & 1 \\ 2 & 2 & 5 & 1 \end{bmatrix} = \frac{1}{6}(-72) = -12$$

$$\text{Volume} = 12$$

$$-4 \begin{bmatrix} 4 & 1 & 1 \\ 5 & 2 & 1 \\ 2 & 5 & 2 \end{bmatrix} + 1 \begin{bmatrix} 0 & 4 & 1 \\ 3 & 5 & 2 \\ 2 & 2 & 5 \end{bmatrix}$$

$$-24 - 48 = -72$$

$$\begin{array}{rrr} -4 & -20 & -5 \\ 4 & 1 & 1 \\ 5 & 2 & 1 \\ 2 & 5 & 2 \end{array}$$

$$+8 + 2 + 25$$

$$\begin{array}{rrr} -10 & 0 & -60 \\ 0 & 4 & 1 \\ 3 & 5 & 2 \\ 2 & 2 & 5 \end{array}$$

$$0 \quad 16 \quad 6$$

Applications of Matrix Operations

- Cryptography

Cryptography

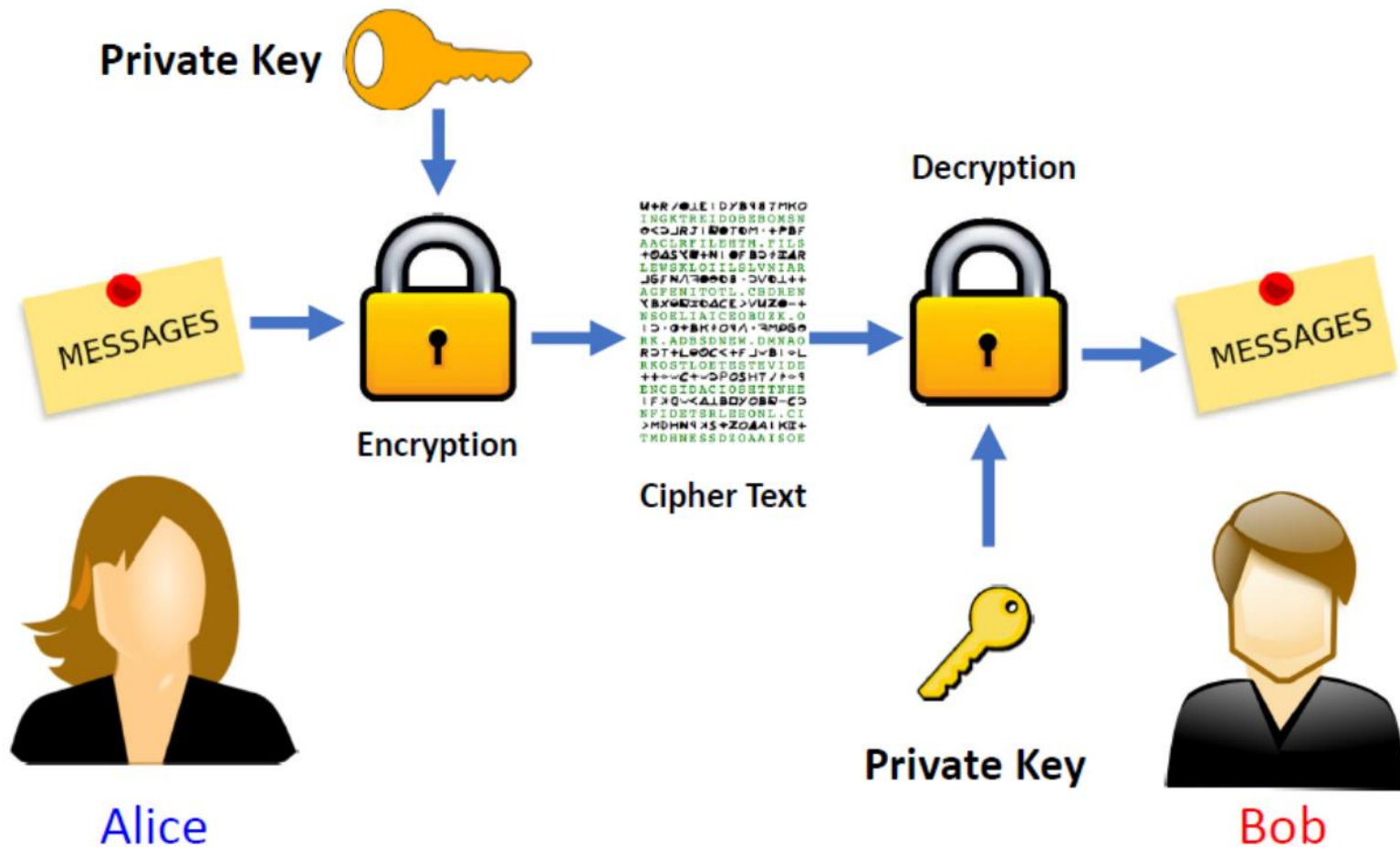


Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

- A **cryptogram** is a message written according to a secret code (the Greek word *kryptos* means “hidden”).
- This section describes a method of using matrix multiplication to **encode** and **decode** messages.

Symmetric Cryptography

Laplace-Transform
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$



Cryptography (cont.)

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

0 = _	14 = N
1 = A	15 = O
2 = B	16 = P
3 = C	17 = Q
4 = D	18 = R
5 = E	19 = S
6 = F	20 = T
7 = G	21 = U
8 = H	22 = V
9 = I	23 = W
10 = J	24 = X
11 = K	25 = Y
12 = L	26 = Z
13 = M	

- Then the message is converted to numbers and partitioned into **uncoded row matrices**, each having n entries.

Forming Uncoded Row Matrices

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Write the uncoded row matrices of size 1×3 for the message MEET ME MONDAY.

$$\begin{bmatrix} 13 & 5 & 5 \end{bmatrix} \begin{bmatrix} 20 & 0 & 13 \end{bmatrix} \begin{bmatrix} 5 & 0 & 13 \end{bmatrix} \begin{bmatrix} 15 & 11 & 4 \end{bmatrix} \begin{bmatrix} 1 & 25 & 0 \end{bmatrix}$$

Cryptography (cont.)



Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

- To **encode** a message, choose an $n \times n$ invertible matrix A and multiply the uncoded row matrices (on the right) by A to obtain **coded row matrices**.

Encoding a Message

Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

Decoded Row Matrix	Encoding Matrix A	Coded Row Matrix
$[13 \quad 5 \quad 5]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [13 \quad -26 \quad 21]$
$[20 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [33 \quad -53 \quad -12]$
$[5 \quad 0 \quad 13]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [18 \quad -23 \quad -42]$
$[15 \quad 14 \quad 4]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [5 \quad -20 \quad 56]$
$[1 \quad 25 \quad 0]$	$\begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$	$= [-24 \quad 23 \quad 77]$

The sequence of coded row matrices is

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77]$$

Finally, removing the brackets produces the cryptogram below.

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77$$

Cryptography (cont.)



Laplace-Trans
 $F(p) = \int_0^\infty e^{-pt} f(t) dt$

- For those who do not know the matrix A , decoding the cryptogram found in the previous example is difficult.
- But for an authorized receiver who knows the matrix A , decoding is simple.
- The receiver need only multiply the coded row matrices by A^{-1} to retrieve the uncoded row matrices.
- In other words, if

$$X = [x_1 \ x_2 \ \cdots \ x_n]$$

is an uncoded $1 \times n$ matrix, then $Y = XA$ is the corresponding encoded matrix.

- The receiver of the encoded matrix can decode Y by multiplying on the right by A^{-1} to obtain

$$YA^{-1} = (XA)A^{-1} = X.$$

Decoding a Message

Laplace-Trans
 $F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$

Use the inverse of the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to decode the cryptogram

13 -26 21 33 -53 -12 18 -23 -42 5 -20 56 -24 23 77.

$$A^{-1} = \frac{1}{\det(A)} \operatorname{Adj}(A) = (1) \begin{bmatrix} -1 & -1 & 0 \\ -10 & -6 & -1 \\ -8 & -5 & -1 \end{bmatrix}$$

Decoding a Message (cont.)

$$[13 \ -26 \ 21] \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [13 \ 5 \ 5]$$

$$[13 \ -53 \ -12] \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [20 \ 0 \ 13]$$

$$[18 \ -23 \ -42] \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [5 \ 0 \ 13]$$

$$[5 \ -20 \ 56] \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [15 \ 14 \ 4]$$

$$[-24 \ 23 \ 77] \begin{bmatrix} -1 & -10 & -8 \\ -1 & -6 & -5 \\ 0 & -1 & -1 \end{bmatrix} = [1 \ 25 \ 0]$$

Q & A

A close-up, slightly blurred image of a document page. The text 'Laplace-Trans' is visible at the top. Below it, the Laplace transform formula is written:
$$F(p) = \int_0^{\infty} e^{-pt} f(t) dt,$$

with the text 'with the condition' partially visible below the formula.